EEPC2201 ELECTRICAL CIRCUIT ANALYSIS LAB (0-0-3)

Sl. No	Name of the Experiments (Any 08 Experiments)	Hrs.
1.	Validation of Network Theorems using AC circuits (Superposition, Thevenin, Norton, Maximum power transfer)	**3**
2.	Study of DC and AC transients for R-L, R-C & R-L-C circuits using digital storage oscilloscope	**3**
3.	Determination of two port network parameters (open circuit and short circuit parameters)	**3**
4.	Determination of two port network parameters(hybrid and transmission parameters).	**3**
5.	Frequency response of low pass and high pass filters.	**3**
6.	Frequency response of band pass and band elimination filters.	**3**
7.	Determination of self-inductance, mutual-inductance and coupling coefficient of a single-phase two winding transformer representing a coupled circuit.	**3**
8.	Study of series and parallel connected magnetically coupled circuits	**3**
9.	Study of resonance in R-L-C series circuit using oscilloscope.	**3**
10.	Study of resonance in R-L-C parallel circuit using oscilloscope.	**3**

Course Outcomes: This course will enable students to:

CO1: validate Superposition, Thevenin, Norton, and Maximum Power Transfer theorems in AC networks.

CO2: interpret the transient responses of R-L, R-C, and R-L-C circuits using a digital storage oscilloscope.

CO3: characterize a given two port network using open circuit, short circuit, hybrid and transmission parameters.

CO4: interpret the frequency response plot of low pass, high pass, band pass, and band elimination filters.

CO5: validate the self-inductance, mutual inductance, and coupling coefficient of a single-phase two winding transformer.

CO6: interpret the response of resonance in R-L-C series and parallel circuits using an oscilloscope

All the above experiments are also to be performed in simulation environment preferably using open-source softwares(MULTISIM, PSIM, PSPICE, LTSPICE etc) and through virtual lab platforms